Exercise session 2

The Plug Flow Reactor



$$\frac{dx_A}{d\left(\frac{V}{F_{A0}}\right)} = r_A\left(x_A, T\right)$$

$$\sum_j \frac{\dot{m}_j\, c_{pj}}{\Omega}\frac{dT}{dz} = r_A\left(x_A, T\right)\left(-\Delta H\right) + U\frac{\pi d_t}{\Omega}\left(T_r - T\right)$$

8 A gas-phase reaction, $A \longrightarrow 2R$, is carried out in a tubular plug flow reactor at $T = 60°C$ and $p_t = 4.75$ bar. The feed consists of 50 mol-% A and 50 mol-% inert at a rate of 4000 kg/h. The molecular weights of A and inert are 40 and 20, respectively. The rate coefficient is $k = 2000\ \text{h}^{-1}$. Determine the reactor size for 35 percent conversion of A.

$X_A = 0{,}35$

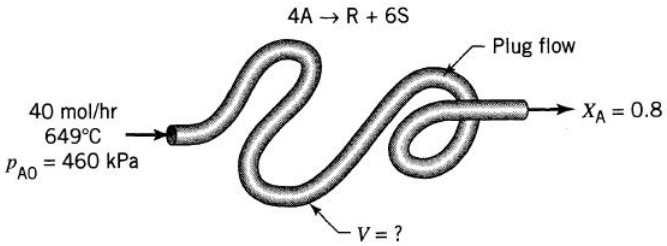
9 The homogeneous gas decomposition of phosphine

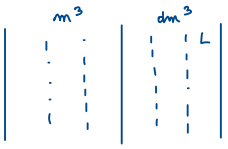
$$4PH_3(g) \longrightarrow P_4(g) + 6H_2$$

proceeds at 649°C with the first-order rate:

$$r_{PH_3} = kC_{PH_3} \qquad \left[\frac{mol}{h.m^3}\right]$$

with $k = 10\ \text{h}^{-1}$. What size of plug flow reactor operating at 649°C and 460 kPa can produce 80% conversion of a feed consisting of 40 mol of pure phosphine per hour?





10 A homogeneous gas reaction $A \longrightarrow 3R$ has a reported rate at 215°C:

$$r_A = 10^{-2} C_A^{1/2} \qquad \left[\frac{mol}{m^3\, s}\right]$$

vérifier unités
si c'était des L à la place des m³, il aurait fallu multiplier par 1000 (au numérateur)

Find the space-time and the space velocity required for 80% conversion of a 50 mol-% A and 50 mol-% inert feed in a plug flow reactor operating at 215°C and 5 atm. To solve this exercise, the following primitive is needed:

$$\int \frac{1+x}{\sqrt{1-x^2}}\, dx = \arcsin x - \sqrt{1-x^2}$$

11 The thermal cracking of propane was studied at atmospheric pressure and 800°C in a tubular reactor with plug flow and operated in the integral mode. The experimental results are given in the table below. The global reaction, $A \longrightarrow 2R$ is considered to be irreversible. First order is assumed so that the rate equation is written:

$$r_A = kC_A$$

The reaction is carried out in the presence of a diluent, stem. (inert !) The dilution ratio is $\kappa = 1$ (moles diluent/moles hydrocarbon). Using the integral method of kinetic analysis, find the value of the rate constant k.

V/F_{A0} (m^3.s/kmol)	$x_A = (F_{A0} - F_A)/F_{A0}$
32	0.488
50	0.63
59	0.685
64	0.714
75	0.76
82	0.782

Additional exercises

12 The process

$$A \xrightarrow{k_1} B \xrightarrow{k_2} C$$

is carried out in a tubular reactor with plug flow. Both reactions are first order. The feed consists of pure A. Given the following data:

$$\begin{aligned} C_{A0} &= 0.05 \text{ kmol/m}^3 \\ F_0' &= 0.15 \text{ m}^3/\text{h} \\ \Omega &= 2 \times 10^{-3} \text{ m}^2 \\ k_1 &= 172.5 \text{ h}^{-1} \end{aligned}$$

calculate the length of the tube to maximize the yield of B in the cases

(a) $k_2 = k_1/2$,

(b) $k_2 = k_1$.

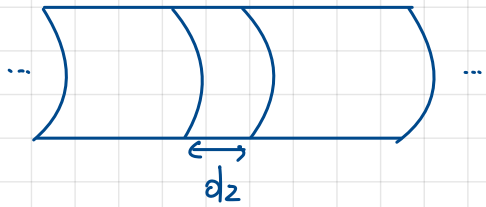
What are the exit concentrations of A, B and C in both cases?

EXERCISE 8

reactor size ? → $V = ?\ m^3$

$A_{in} - A_{out} = A_{accum} + A_{reagit}$

$A_{accum} = 0$ car plus flow → on considère que rien ne s'accumule (flux continu)



$F_{A_{in}} - F_{A_{out}} = 0 + r_A \cdot dV$

$-\Delta F_A = r_A\, dV$

$-dF_A = r_A\, dV$ et $F_A = F_{A_0} \cdot (1 - X_A)$

$\boxed{F_{A_0}\, dX_A = r_A \cdot dV}$ $\quad dF_A = -F_{A_0}\, dX_A$

$dV = \dfrac{F_{A_0} dX_A}{r_A}$ or, je sais que $r_A = k \cdot C_A$

$dV = \dfrac{F_{A_0}}{k \cdot C_A}\, dX_A$

$$\frac{dx_A}{d\left(\frac{V}{F_{A0}}\right)} = r_A(x_A, T)$$

→ on peut le sortir car cst

↳ $F = 4000$ kg/h

The feed consists of 50 mol-% A and 50 mol-% inert

↳ pourcentage molaire ! => d'abord passer en flow molaire

$M_{Tot} = 50\% \cdot 40 + 50\% \cdot 20$

$= 20 + 10 = 30$ kg/mol

$F = \dfrac{4000 \text{ kg/h}}{30 \text{ kg/mol}} = 133{,}33 \dfrac{\text{mol}}{\text{h}}$

$F_{A_0} = 50\% \cdot 133{,}33 = 66{,}66 \dfrac{\text{mol}}{\text{h}}$

↳ C_A ? → exprimer en fct° de X_A parce que la concentration évolue au cours de la réaction

$P \cdot V = n \cdot R\, T$

$\dfrac{P}{RT} = \dfrac{n}{V} = C$

$C_A = \dfrac{P_A}{R \cdot T} = \dfrac{y_A \cdot P_T}{R \cdot T}$

Pour trouver y_A on fait un tableau d'avancement :

	A	I	$\longrightarrow$	2 R	TOTAL
z_0	F_{A0}	$F_{I_0} = F_{A0}$		0	$2\,F_{A0}$
z	$F_{A0}.(1-X_A)$	F_{A0}		$2\,F_{A0}.X_A$	$F_{A0}.(1-X_A) + F_{A0} + 2F_{A0}.X_A$ $\Leftrightarrow F_{A0} - F_{A0}X_A + F_{A0} + 2F_{A0}X_A$ $\Leftrightarrow 2F_{A0} + F_{A0}X_A$ $\Leftrightarrow F_{A0}.(2+X_A)$

$$\longrightarrow \quad y_A = \frac{\text{flux de A}}{\text{flux TOTAL}}$$

$$= \frac{F_{A0}.(1-X_A)}{F_{A0}.(2+X_A)}$$

$$= \frac{1-X_A}{2+X_A}$$

On peut tout remettre dans la formule du dV :

$$dV = \frac{F_{A0}}{R.C_A}\,dX_A \quad \Leftrightarrow \quad dV = \frac{F_{A0}}{R.\frac{\frac{1-X_A}{2+X_A}.P_T}{R.T}}\,dX_A$$

$$\Leftrightarrow \quad dV = F_{A0}.RT.\frac{1}{R}.\frac{2+X_A}{1-X_A}.\frac{1}{P_T}\,dX_A$$

$\longrightarrow$ il faut intégrer le tout pour trouver une expression de V :

On va d'abord faire un "artifice de calcul" afin de faire apparaître $(1-X_A)$ au numérateur :

$$\Leftrightarrow \quad dV = F_{A0}.R.T.\frac{1}{R}.\frac{(1-X_A).(-1)+3}{1-X_A}.\frac{1}{P_T}\,dX_A$$

$$\Leftrightarrow \quad \int dV = \frac{F_{A0}.R.T}{R.P_T}.\int \frac{(1-X_A).(-1)}{1-X_A} + \frac{3}{1-X_A}\,dX_A$$

$$\Leftrightarrow \quad V = \frac{F_{A0}.R.T}{R.P_T}.\left(-1.[X_A]_0^{0,35} + 3.[\ln(1-X_A)]_0^{0,35}.(-1)\right)$$

$$= \frac{66,66.8,31.(60+273)}{2000.4,75.10^5}.\left(-0,35 - 3.\ln(1-0,35)\right)$$

$$= 1,82.10^{-4}\ m^3 = 0,182\ L$$

EXERCICE 3

What size → $V = ?\ m^3$

Plug flow reactor: $F_{A_0} dX_A = r_A \cdot dV$

$\Leftrightarrow dV = \frac{F_{A_0}}{r_A} dX_A$ where r_A is a function of X_A

$\Leftrightarrow dV = F_{A_0} \cdot \frac{1}{k \cdot C_A} dX_A$ bc $r_A = k C_A^n$ and here it's a first order reaction

From the ideal gas law : $P V = n \cdot R \cdot T$

$\Leftrightarrow \frac{n_A}{V} = \frac{P_A}{R \cdot T} = C_A$

$\Leftrightarrow dV = F_{A_0} \cdot \frac{1}{k} \cdot \frac{RT}{P_A} \cdot dX_A$

$= F_{A_0} \cdot \frac{1}{k} \cdot \frac{RT}{y_A \cdot P_T} \cdot dX_A$

To find y_A, we need to do a progress table as it evolves as a function of the progress of the reaction :

	A		R	S	TOTAL
	4 PH_3	→	P_4	6 H_2	
z_0	$4 F_{A_0}$		0	0	$4 F_{A_0}$
z	$4 F_{A_0}(1-X_A)$		$F_{A_0} X_A$	$6 F_{A_0} X_A$	$4F_{A_0} - 4F_{A_0}X_A + F_{A_0}X_A + 6F_{A_0}X_A$
					$= 4F_{A_0} + 3F_{A_0}X_A$
					$= F_{A_0}(4 + 3X_A)$

pq je mets un X_A à l'inerte ? → parce que c'est pas un inerte mais un produit de la décomposition

↳ $y_A = \frac{\#\ mol\ of\ A}{\#\ mol\ TOT} = \frac{4 F_{A_0}(1-X_A)}{F_{A_0}(4+3X_A)}$

$= \frac{4(1-X_A)}{4+3X_A}$

$\Leftrightarrow \int dV = \int F_{A_0} \cdot \frac{1}{k} \cdot \frac{RT}{P_T} \cdot \frac{4+3X_A}{4(1-X_A)} \cdot dX_A$

$\Leftrightarrow V = \frac{F_{A_0} \cdot R \cdot T}{k \cdot P_T} \cdot \int_0^{0,8} \frac{(1-X_A)(-3)+7}{4(1-X_A)} dX_A$

$$\Leftrightarrow \quad V = \frac{F_{A0} \cdot R \cdot T}{k \cdot \uparrow T} \cdot \int_0^{0,8} \frac{(1 - X_A) \cdot (-3)}{4 \cdot (1 - X_A)} + \frac{7}{4 - 4X_A} \, dX_A$$

$$\Leftrightarrow \quad V = \frac{F_{A0} \cdot R \cdot T}{k \cdot \uparrow T} \cdot \int_0^{0,8} \frac{-3}{4} + 7 \cdot \frac{1}{4 - 4X_A} \, dX_A$$

$$= \frac{F_{A0} \cdot R \cdot T}{k \cdot \uparrow T} \cdot \left(\frac{-3}{4} \cdot [X_A]_0^{0,8} + 7 \cdot \left[\ln(4 - 4X_A) \right]_0^{0,8} \frac{1}{-4} \right)$$

$$= \frac{F_{A0} \cdot R \cdot T}{k \cdot \uparrow T} \cdot \left(\frac{-3}{4} \cdot (0,8 - 0) \; \frac{-7}{4} \ln\left(\frac{4 - 4 \cdot 0,8}{4 - 4 \cdot 0} \right) \right)$$

$$= \frac{F_{A0} \cdot R \cdot T}{k \cdot \uparrow T} \cdot \left(\frac{-2,4}{4} - \frac{7}{4} \ln\left(\frac{0,8}{4} \right) \right)$$

$$= \frac{40 \cdot 8,31 \cdot (649 + 273)}{10 \cdot 460 \cdot 10^3} \cdot \left(\frac{-2,4}{4} - \frac{7}{4} \ln\left(\frac{0,8}{4} \right) \right)$$

$$= 0,1476 \; m^3$$

EXERCISE 10

Pour un plug flow : space-time = residence time pour un liquide incompressible ici on est avec un gaz → ça ne s'applique pas.

Space time $= C_{A_0} . V . \frac{1}{F_{A_0}}$

↳ $C_{A_0} = y_{A_0} . \frac{P_T}{R.T} = 0,5 . \frac{506,625}{8,31 . (215+273)} = 0,0624643 \ \frac{mol}{m^3}$ (unités SI)

↳ $\frac{V}{F_{A_0}}$? $\quad dV = \frac{F_{A_0} dx_A}{r_A}$

$\Leftrightarrow \frac{dV}{F_{A_0}} = \frac{dx_A}{r_A} = \frac{dx_A}{10^{-2} C_A^{1/2}}$

Il faut intégrer pour avoir un V et non un dV. Le problème c'est que C_A est fonction de x_A → exprimer avec y_A et le tableau d'avancement en partant de la loi des gaz parfaits pour faire apparaître le x_A :

	A	I → 3 R	TOTAL
0	F_{A_0}	F_{A_0} 0	$2 F_{A_0}$
t	$F_{A_0} . (1-x_A)$	F_{A_0} $3F_{A_0} . x_A$	$F_{A_0} . (1-x_A+1) + 3F_{A_0} . x_A$ $= F_{A_0} . (2-x_A) + 3F_{A_0} . x_A$

$$y_A = \frac{A}{TOT} = \frac{F_{A_0} . (1-x_A)}{F_{A_0} . (2-x_A) + 3F_{A_0} . x_A}$$

$$= \frac{1-x_A}{2-x_A+3x_A}$$

$$= \frac{1-x_A}{2+2x_A}$$

→ on remplace mnt l'expression de C_A :

$C_A = y_A . \frac{P_T}{R.T}$ (via $P.V = n.R.T$)

$= \frac{1-x_A}{2.(1+x_A)} . \frac{P_T}{R.T}$

On peut mnt intégrer notre expression de base puisqu'on a réussi à l'exprimer en fonction de x_A :

$$\frac{dV}{F_{A_0}} = \frac{dx_A}{10^{-2}\, C_A^{1/2}}$$

$$\Leftrightarrow \frac{dV}{F_{A_0}} = \frac{dx_A}{10^{-2}} \left(\frac{2.(1+x_A)}{1-x_A} \cdot \frac{R.T}{1T} \right)^{1/2}$$

Now we have to integrate... To facilitate the work, we multiply the numerator and denominator by $\sqrt{1-x^2}$

$$\Leftrightarrow \frac{dV}{F_{A_0}} = \frac{dx_A}{10^{-2}} \cdot \sqrt{\frac{2.(1+x_A)}{1-x_A} \cdot \frac{RT}{1T}} \cdot \frac{\sqrt{1+x_A}}{\sqrt{1+x_1}}$$

$$= 10^2 \cdot \sqrt{2} \cdot \sqrt{\frac{RT}{1T}} \cdot \frac{\sqrt{1+x_A}}{\sqrt{1-x_A}} \cdot \frac{\sqrt{1+x_A}}{\sqrt{1+x_1}} \, dx_1$$

produits remarquables

Remarquable !

$(a+b)^2 = a^2 + 2ab + b^2$
$(a-b)^2 = a^2 - 2ab + b^2$
$(a+b)(a-b) = a^2 - b^2$

$$= 10^2 \cdot \sqrt{2} \cdot \sqrt{\frac{RT}{1T}} \cdot \frac{1+x_A}{\sqrt{1-x_A^2}} \, dx_1$$

$$\Leftrightarrow \int \frac{1}{F_{A_0}} dV = \int 10^2 \cdot \sqrt{2} \cdot \sqrt{\frac{RT}{1T}} \cdot \frac{1+x_A}{\sqrt{1-x_A^2}} \, dx_1$$

and from the instructions we know that : $\int \frac{1+x}{\sqrt{1-x^2}}\,dx = \arcsin x - \sqrt{1-x^2}$

$$\Leftrightarrow \frac{V}{F_{A_0}} = 10^2 \sqrt{2} \cdot \sqrt{\frac{RT}{1T}} \cdot \left[\arcsin x_1 - \sqrt{1-x_1^2} \right]_0^{x_{1f}=0,8}$$

$$\Leftrightarrow \frac{V}{F_{A_0}} = 10^2 \cdot \sqrt{2} \cdot \sqrt{\frac{8,31 \cdot (215+273)}{506,625}} \cdot \left[\arcsin 0,8 - \sqrt{1-0,8^2} - \arcsin 0 + \sqrt{1-0^2} \right]$$

↑ 1 atm = 101 325 Pa

$$\Leftrightarrow \frac{V}{F_{A_0}} = 400,11249 \cdot 1,3273$$

$$\Leftrightarrow \frac{V}{F_{A_0}} = 531,0693 \left[m^3 \cdot \frac{s}{mol} \right]$$

$$\Rightarrow \text{space-time} = \tau = \frac{C_{A_0} \cdot V}{F_{1_0}} = 0,0624643 \cdot 531,0693 \quad \frac{\cancel{mol}}{\cancel{m^3}} \cdot \cancel{m^3} \cdot \frac{s}{\cancel{mol}}$$

$$= 33,17 \; s$$

EXERCISE 11

We are looking for the value of the rate constant k : let's isolate it in the general equation of a plug flow

$$F_{A_0} dX_A = r_A . dV$$

$$\Leftrightarrow F_{A_0} dX_A = k . C_A . dV$$

$$\Leftrightarrow k = \frac{F_{A_0} . dX_A}{C_A . dV}$$

→ faut d'abord intégrer mais 1) C_A est dépendant de la pression partielle en A qui s'exprime selon la fraction molaire multiplié par la pression totale. Et cette fraction molaire en A évolue au cours de la réaction, autrement dit, elle est fonction de X_A ...

$$\Leftrightarrow k . dV = F_{A_0} . \frac{RT}{P_A} dX_A \qquad P . V = n . R . T \Leftrightarrow \frac{n}{V} = \frac{P_A}{RT} = C_A$$

$$\Leftrightarrow k . dV = F_{A_0} . \frac{R . T}{y_A P_T} dX_A$$

or, $y_A = \frac{\text{\# mol A}}{\text{\# mol TOT}}$ → tableau d'avancement

inert here is steam

	A	$\longrightarrow$	$2R$	$+$	I	TOTAL
Z_0	F_{A_0}		0		F_{A_0}	F_{A_0}
Z	$F_{A_0} . (1-X_A)$		$2F_{A_0} . X_A$		F_{A_0}	$F_{A_0} . (1-X_A+2X_A+1)$ $= F_{A_0} . (2+X_A)$

$$\rightarrow y_A = \frac{F_{A_0} . (1-X_A)}{F_{A_0} . (2+X_A)} = \frac{1-X_A}{2+X_A}$$

$$\Leftrightarrow k . dV = F_{A_0} . \frac{R . T}{P_T} . \frac{2+X_A}{1-X_A} dX_A$$

$$\Leftrightarrow \int k \, dV = \frac{F_{A_0} . R . T}{P_T} . \int_0^{X_A} \frac{(1-\tilde{X}_A) . (-1) + 3}{1-\tilde{X}_A} d\tilde{X}_A$$

$$\Leftrightarrow k . V = \frac{F_{A_0} . R . T}{P_T} . \int_0^{X_A} \frac{(1-\tilde{X}_A) . (-1)}{1-\tilde{X}_A} + \frac{3}{1-\tilde{X}_A} d\tilde{X}_A$$

$$\Leftrightarrow k . V = \frac{F_{A_0} . R . T}{P_T} . \left(\left[-\tilde{X}_A \right]_0^{X_A} + 3 . \left[\ln (1-\tilde{X}_A) \right]_0^{X_A} . (-1) \right)$$

$$\Leftrightarrow k . V = \frac{F_{A_0} . R . T}{P_T} . \left(-X_A - 3 . \ln (1-X_A) \right)$$

$$\Leftrightarrow \quad k = \frac{F_{A0} \cdot R \cdot T}{P_T \cdot V} \cdot \left(-X_A - 3 \ln (1 - X_A) \right)$$

$$\Leftrightarrow \quad k = \left(\frac{V}{F_{A0}} \right)^{-1} \cdot \frac{R \cdot T}{P_T} \cdot \left[-X_A - 3 \ln (1 - X_A) \right] \qquad [s^{-1}]$$

Compute everything in an excel :

V/F_{A0} ($m^3.s/kmol$)	$x_A = (F_{A0} - F_A)/F_{A0}$	k
32	0.488	4,18
50	0.63	4,14
59	0.685	4,14
64	0.714	4,18
75	0.76	4,13
82	0.782	4,06

↳ ⚠ à bien ÷ par 1 000 pour être en mol et non en kmol !

average : 4,14 s^{-1}